Sampling, FFT, and the Pre-Lab1 notebook

Fall 2026 · Prof. Ehsan Roohi

Lecture and guided Jupyter walkthrough

Learning goals

- Explain sample rate, sample interval, record length, and Nyquist frequency
- Run prepared analysis cells in a Jupyter notebook
- Interpret time histories, RMS values, and one-sided FFT spectra
- Recognize aliasing before collecting Lab 1 sound data

Connection to Lab 1

PRE-LAB

A known voltage signal contains two pure tones.

Amplitude describes signal size; frequency describes how fast it repeats; phase sets its starting position.

LABORATORY

A microphone converts changing sound pressure into changing voltage.

Moku:Go stores voltage samples. The waveform shows change in time; the spectrum shows the frequencies present.

Jupyter notebook structure

MARKDOWN CELLS

Instructions, equations, and response areas.

Double-click to edit. Use Shift+Enter to render and move forward.

CODE CELLS

Python commands that define variables, calculate results, and make figures.

Run from top to bottom with Shift+Enter.

Opening and saving the notebook

- Download the .ipynb file from Canvas or open it from the course website in Colab
- Save a personal copy before editing
- Run cells from top to bottom
- Keep code outputs and figures in the submitted file

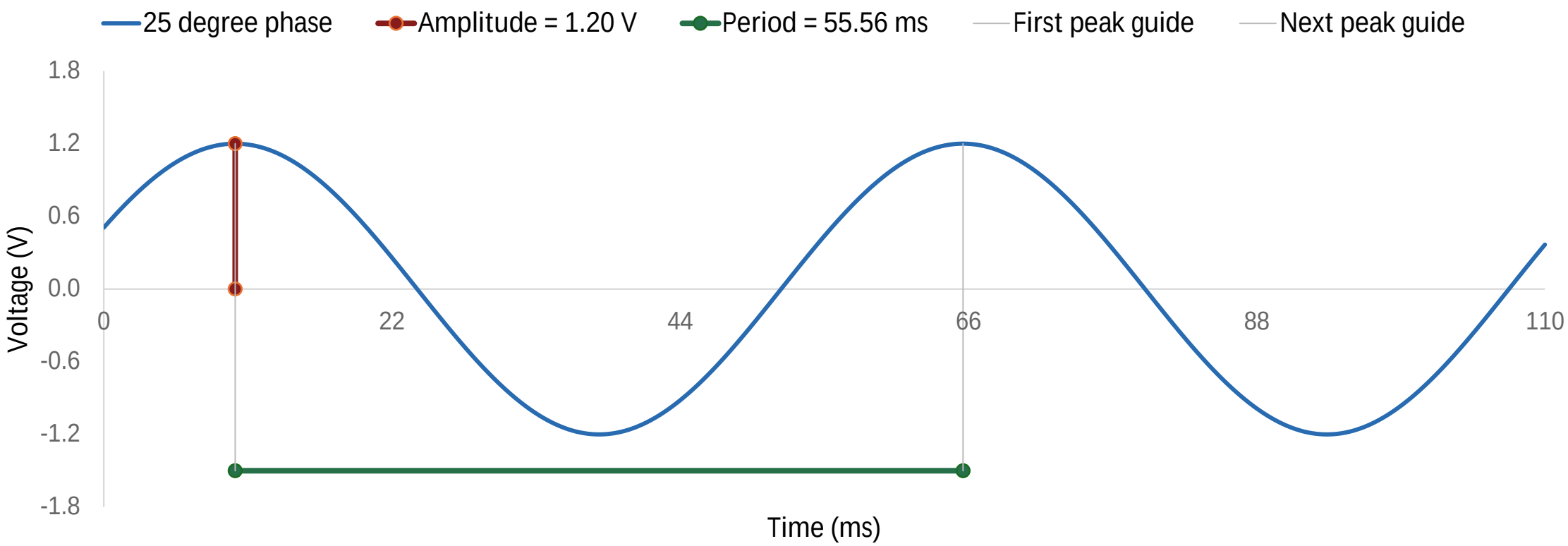
Execution order matters

```
import numpy as np
A = 1.20
print(A)

# A later cell can use A only after this cell runs.
rms_component = A / np.sqrt(2)
print(rms_component)
```

A restart clears stored variables. Run All rebuilds the notebook state from the first cell.

A sinusoid: what the three parameters mean



A s= 1.20V from zero to peak. Ts= 55.56ms between consecutive peaks.

xs(t) = As sin(2¯Ťsű Ь ›s), with fs = 18 I Î I·yŘ ›s = 25°

The new two-tone signal

```
A1, f1, phi1_deg = 1.20, 18.0, 25.0
A2, f2, phi2_deg = 0.45, 42.0, -15.0

def signal(t):
    return (A1*np.sin(2*np.pi*f1*t + np.deg2rad(phi1_deg))
            + A2*np.cos(2*np.pi*f2*t + np.deg2rad(phi2_deg)))
```

Subscript 1 means tone 1; subscript 2 means tone 2. A is in volts, f is in hertz, and phi_deg is in degrees.

Phase relative to a zero-phase sine wave

The prepared sampling analysis

```
for fs in sample_rates:
    n = int(round(duration*fs))
    t = np.arange(n)/fs
    x = signal(t)
    records[fs] = (t, x)
    sample_mean, sample_rms = mean_and_rms(x)
    print(fs, sample_mean, sample_rms)
```

Students transfer the printed values to an analysis table and calculate RMS percent error.

Reading the time-domain figures

- Compare sample markers with the dense reference over the same time window
- Look for too few points per cycle and apparent changes in shape
- Read the printed mean and RMS values
- Avoid judging sample quality from smooth connecting lines

FFT peaks, windowing, and amplitude

READING THE SPECTRUM

Horizontal axis: frequency (Hz).

Peak location identifies a tone.

Peak height estimates amplitude (V).

One-sided means nonnegative frequencies for a real-valued signal.

WHY USE A WINDOW?